

R22

Code No: 185BB

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech III Year I Semester Examinations, January – 2025

COMPUTER NETWORKS

(Common to CSE, CE(SE), CSE(AI&ML;), CSE(DS), AI&DS;, AI&ML;, CSD)

Time: 3

Hours

Max.

Marks: 60

Note: This question paper contains two parts A and B.

i) Part-A for 10 marks, ii) Part-B for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of ten questions (numbered from 2 to 11) carrying 10 marks each. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART – A

(10 Marks)

1.a) What is Coaxial Cable?

[1]

b) List out the two sub layers of data link layer.

[1]

c) What is the purpose of Aloha protocols?

[1]

d) What is channel allocation problem?

[1]

e) Define congestion in a network.

[1]

f) What is Interworking?

[1]

g) What is UDP?

[1]

h) What is process to process delivery?

[1]

i) Define WWW.

[1]

j) What is MPEG?

[1]

PART – B

(50 Marks)

2. Explain in detail about TCP/IP network reference model with neat diagram.

[10]

OR

3.a) Describe the functionalities provided by data link layer.

b) Explain fiber optic communication in detail.

[5+5]

4. Describe sliding window protocol using selective repeat mechanism.

[10]

OR

5. Explain the concept of data link layer switching in detail.

[10]

6.a) Explain the design issues in network layer.

b) Explain congestion control mechanisms at network layer.

[5+5]

OR

7.a) Describe the multicast routing in detail.

b) Discuss about network layer in the Internet.

[5+5]

8.a) Describe the functionalities of transport layer.

b) Give a brief note on connection management at transport layer.

[5+5]

OR

9.a) With a neat architecture, explain TCP in detail.

b) Enumerate the differences between TCP and UDP.

[5+5]

10.a) Explain in detail about HTTP operation.

b) Summarize the functions performed by DNS.

[5+5]

OR

11.a) Describe the message format, message transfer and the underlying protocol involved in the working of the electronic mail.

b) Give a short note on audio streaming at application layer.

[5+5]

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